Ritam

PARTICIPANT HANDBOOK · ADVANCED COURSE

The Mind in the Light of Modern and Spiritual Sciences

MODULE 3

The Physiology of the Mind: Wave-Equation

- 3.1 Vṛtti: The Waves of the Mind
- 3.2 Interference: How the Knower Meets the Known
- 3.3 Saṁskāra as Live Filter
- 3.4 Signal and Noise: Why the Mind Distorts
- 3.5 Attention and the Construction of Experience
- 3.6 Resonance, Rhythm and Coherence
- 3.7 Body, Prāṇa and the Energetics of Experience
- 3.8 Two Modes of Knowing: Mind as Bridge Between Domains

Notation and terms used in Module 3

The symbols and terms the eight lessons use, with the lessons' own wording and the lesson where each first appears. Use it alongside the lesson pages.

Term	In the lessons' words	First appears
vṛtti	A whirling or wave-formation in the substance of the mind; in its precise sense (3.2), the whole resultant of the meeting.	3.1
ψ(t); Ψ(f)	One wave seen in the temporal domain (unfolding in time) and the spectral domain (as a function of frequency); equivalent via the Fourier transform.	3.1
$\psi_x(t)$	The object-wave: what is actually present, with its own intrinsic frequency, which the lessons write f_s.	3.1 (defined 3.2)
$\psi_m(t)$	The attention wave: what you bring, with its own frequency f_m, which manas sets. In 3.5: $\psi_m(t) = A_m \cdot \cos(2\pi f_m \cdot t + \varphi_m)$.	3.1 (defined 3.2, 3.5)
superposition; ψexp(t)	The two waves added in the one field where both arise, chitta; ψexp(t) is their resultant, the experience you live.	3.1 (explained 3.2)
carrier; envelope; beat; Δf	Carrier: at the mean frequency $(f_s + f_m)/2$, the content layer. Envelope: the clarity layer; its felt pulse, the beat, is $\Delta f = \lvert f_s - f_m \rvert$, slowing and vanishing as $\Delta f \to 0$.	3.1 (defined 3.2)
Δφ; φ_m	Δφ: the standing phase-offset the attention wave carries into every meeting, the ahaṁkāra. Buddhi corrects it, adjusting φ_m, the attention wave's phase.	3.2
h(t)	The impulse response: how your system has learned to respond, formed by the saṁskāras held in chitta.	3.2 (defined 3.3)
∗ (convolution)	Not simple multiplication: the incoming signal combined with the memory of all previous responses to similar signals.	3.2 (explained 3.3)
H(f)	The transfer function: the same conditioning by frequency. In the coherent state it approaches 1 (3.6).	3.3
n(t); noise floor; SNR	All unwanted noise; the noise floor, below which no signal reaches consciousness. SNR = P_signal / P_noise.	3.2 (defined 3.4)
the three malas	Āṇava, the distorted standing tilt; māyīya, buddhi's miscalibrated correction; kārma, past conditioning keeping h(t) ringing. Not moral failures.	3.4
witness (sākṣī)	The awareness observing all experience all along: not a new object; you recognise yourself as the witness.	3.4
G(t); A_m	The gain: the strength with which the body and prāṇa drive the attention wave; $G(t) \cdot A_m$ is the actual strength that reaches the meeting.	3.2 (defined 3.5)
BW	The bandwidth of attention: $BW \cdot \Delta t \geq$ constant.	3.5
ε; w_1, w_2	Prediction error $\varepsilon(t) = \psi_x(t) - \psi_predicted(t)$; the weights on prediction and on error.	3.5
resonance; coherence; Q	Frequency match, maximum energy transfer; parts in phase; Q, how selective a resonance is.	3.6
reference rhythm	A stable rhythm to synchronize with: in 3.7, slow breath.	3.4 (developed 3.7)
the five prāṇas	Prāṇa (intake), apāna (release), samāna (assimilation), udāna (expression), vyāna (circulation).	3.7
aparā vidyā; parā vidyā	Difference-based knowing, $\Delta f > 0$; unity-based knowing, the perfected resultant ($\Delta f \to 0$), the knowing by which the Imperishable is attained (Muṇḍaka 1.1.5).	3.1, 3.2 (developed 3.8)
Self-Recognition	Not a component of the experience equation: the knowing-by-the-Self of the Self, beyond even the perfected resultant.	3.1 (explained 3.2)
saṁyama	Dhāraṇā, dhyāna and samādhi as one practice reducing the beat.	3.8

Plain words. **Amplitude**: the height of the wave. **Frequency**: how fast the wave oscillates, complete cycles in one unit of time (usually one second); **Hz** is the unit, as in 329 Hz. **Fourier transform**: converts the wave from the temporal to the spectral domain. **Linear zone** (3.2): where the medium responds in proportion to what it receives, so superposition holds. **Bandwidth BW**: the range of frequencies the mind spans at a given moment.

Two notes. 3.1 writes sine, 3.5 cosine; they differ only by a shift in phase. Lessons 3.2 to 3.5 write the combination with + signs; the handbook follows each lesson's own tracker line.

The module's equation (as Lessons 3.6, §13, and 3.7 state it): $\psi exp(t) = [\psi_x(t) * h(t)] \cdot G(t) \cdot \psi_m(t) + n(t)$

Lesson 3.1: Vṛtti: The Waves of the Mind

Aim: to understand vṛtti, the wave-formations that constitute all experience, and to begin to see your own experience not as solid, permanent fact but as a pattern of formations that can change. **Online:** Practice page and understanding check for 3.1 (vedanta.nz, My Journey).

1 In my own words

In three sentences, say what a vṛtti is, and what the lesson means when it says the wave is not separate from the water.

2 Terms to recall

Write each in your own words, then check the key points at the back.

Term	In my own words
Vṛtti	
Citta-vṛtti-nirodha	
Temporal domain ψ(t)	
Spectral domain Ψ(f)	
Fourier transform	
Viparyaya	
Vikalpa	
Phase φ	

3 Fill the gaps

Viparyaya feels exactly like _____ while it is happening.

Manas sets the _____ of attention; the body and prāṇa set the amplitude.

Switch from the temporal domain to the spectral domain and you lose _____.

4 Spot the slip

Each statement contains a mistake. Correct it in the lines below.

"Yoga aims at a blank mind: the waves have to be destroyed before the Self can appear."

"In deep sleep there is no vṛtti at all; sleep lies below the waves of the mind."

Lesson 3.1: Vṛtti: The Waves of the Mind (continued)

5 Draw it in your own way

Figure 3.1.d. The lesson's caption reads: “The anatomy of a wave: amplitude set by the body and prāṇa, frequency by manas, phase by ahaṁkāra's standing tilt and buddhi's correction, and duration shaped by the saṁskāras held in chitta.” Draw one vṛtti as a wave. Mark its amplitude, one full cycle, and where it rises from and subsides back to the baseline. At the foot of the frame write what the lesson says sets each parameter.

One vṛtti as a wave

Amplitude A set by:

Frequency f set by:

Phase φ shaped by:

Duration shaped by:

6 Apply

Late at night Mere keeps picturing a meeting next week going badly, and says to herself, “This is just who I am: a worrier.” Using the lesson's five types of vṛtti, say which kind this is, and what the lesson's shift “from identity to description” would invite her to ask, without denying that the worry feels real.

7 Tracker

Module 3 builds one equation, lesson by lesson, starting from ψexp(t) = ? In the lesson's own notation, write what Lesson 3.1 adds, then copy the line the lesson's tracker gives.

8 Check my notes

Key points for 3.1 are on the page marked 3.1 at the back of the handbook. There are no marks and no self-rating: compare your notes and decide for yourself.

Lesson 3.2: Interference: How the Knower Meets the Known

Aim: to recognise that perception is never passive reception but an active meeting between what is actually there and what you bring to it. **Online:** Practice page and understanding check for 3.2 (vedanta.nz, My Journey).

1 In my own words

In three sentences, say what happens when the object-wave and the attention wave meet, and what the carrier and the envelope each give you.

2 Terms to recall

Write each in your own words, then check the key points at the back.

Term	In my own words
Superposition	
Object-wave $\psi_x(t)$	
Attention wave $\psi_m(t)$	
Carrier	
Envelope	
The beat, $\lvert f_s - f_m \rvert$	
Standing phase-offset $\Delta\varphi$	
Adhyāsa	

3 Fill the gaps

Waves can only superpose in a shared _____.

The carrier oscillates at the _____ frequency, so the content of experience is co-created.

With matched frequencies, the factor $\cos(\Delta\varphi/2)$ scales the _____ of the experience.

4 Spot the slip

Each statement contains a mistake. Correct it in the lines below.

"When the two waves meet, they merge into a new third wave, and the two originals are gone."

"I did not hear the clock strike, so for me the sound was never there."

Lesson 3.2: Interference (continued)

5 Draw it in your own way

Figure 3.2.a. The lesson's caption reads: “Every perception is two waves meeting: object-wave $\psi_x(t)$ plus attention-wave $\psi_m(t)$ give a resultant (the vṛtti) with a carrier (content) and an envelope (clarity).” Draw two waves of slightly different frequency, one in each of the first two strips, and their sum in the third. On the resultant mark the carrier, the envelope and one beat.

$\psi_x(t)$ the object-wave

$\psi_m(t)$ the attention wave

The resultant (the vṛtti): mark the carrier, the envelope and one beat

6 Apply

At choir practice the director gives Wiremu a short note on his timing. Expecting to be criticised, he hears a reprimand, and the rest of the rehearsal feels patchy and effortful. In the lesson's terms, say what ψ_x and ψ_m are here, what Δf and $\Delta\varphi$ might describe, and one thing the lesson suggests he could do before the next rehearsal.

7 Tracker

The module's equation began in 3.1 as a sum of vṛttis. In the lesson's own notation, write what Lesson 3.2 adds: the meeting, the two layers, Δf and $\Delta\varphi$, and what full alignment gives. End with the equation the lesson's tracker gives.

8 Check my notes

Key points for 3.2 are on the page marked 3.2 at the back of the handbook. There are no marks and no self-rating: compare your notes and decide for yourself.

Lesson 3.3: Saṁskāra as Live Filter

Aim: to understand saṁskāras not as static memory traces but as active, real-time filters that shape every incoming perception before you are even aware of it. **Online:** Practice page and understanding check for 3.3 (vedanta.nz, My Journey).

1 In my own words

In three sentences, say how saṁskāras act on an incoming signal, and why two people hear the same remark differently.

2 Terms to recall

Write each in your own words, then check the key points at the back.

Term	In my own words
Impulse response h(t)	
Settling time	
Overshoot	
Residual ripple	
Convolution, $\psi_x(t) * h(t)$	
Transfer function H(f)	
Damping	
Reference beam (hologram)	

3 Fill the gaps

In a well-damped mind you feel your feelings fully, but the response is _____, and you settle relatively quickly.

In the frequency domain, convolution becomes simple _____: Output(f) = Input(f) · H(f).

To transform a memory, you must change the _____ beam.

4 Spot the slip

Each statement contains a mistake. Correct it in the lines below.

"My reaction depends only on what was said to me."

"A saṁskāra is stored at the time and place of the event, like a file."

Lesson 3.3: Saṁskāra as Live Filter (continued)

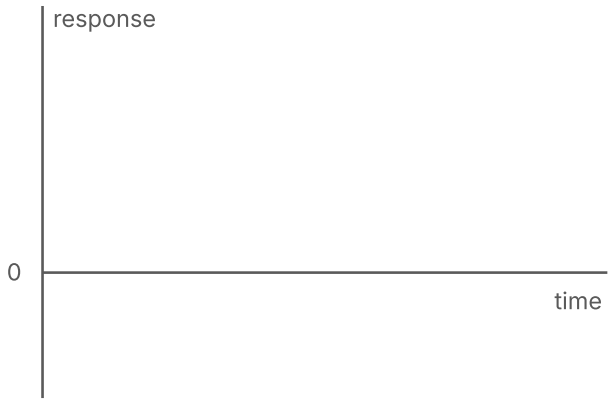
5 Draw it in your own way

Figure 3.3.b. The lesson's caption reads: “Two ways the mind settles: a well-damped response rises and falls once, while an under-damped response overshoots and keeps ringing.” On each pair of axes sketch one response to the same single input. Mark the settling time on both, and the overshoot and residual ripple where they appear.

Well-damped



Under-damped



6 Apply

Priya's manager says, “Can we talk about last week's report?” Without describing yourself, say how two different h(t) could shape what arrives for Priya, and which of the dampers the lesson names (ethics and integrity, viveka, breath regulation, a calm environment and unhurried time, seeing one's own pattern) might help.

7 Tracker

In 3.2 the equation was $\psi exp(t) = \psi_x(t) + \psi_m(t)$. In the lesson's own notation, write what Lesson 3.3 adds and what it does to the incoming signal before attention meets it. End with the equation as the lesson gives it.

8 Check my notes

Key points for 3.3 are on the page marked 3.3 at the back of the handbook. There are no marks and no self-rating: compare your notes and decide for yourself.

Lesson 3.4: Signal and Noise: Why the Mind Distorts

Aim: to understand mental noise, unwanted fluctuation that obscures clarity, and to see the distortions in perception not as signs of personal failure but as reflections of a system with a high noise floor. **Online:** Practice page and understanding check for 3.4 (vedanta.nz, My Journey).

1 In my own words

In three sentences, say what the noise floor is, and why the lesson says a high noise floor is not a moral problem.

2 Terms to recall

Write each in your own words, then check the key points at the back.

Term	In my own words
Noise	
Signal-to-noise ratio (SNR)	
Noise floor	
Āṇava mala	
Māyīya mala	
Kārma mala	
Antarāyas	
Eka-tattva-abhyāsa	

3 Fill the gaps

SNR = 20 dB means the signal is one _____ times stronger than the noise.

In signal-processing terms, kārma mala corresponds to a long _____.

Practice lowers the noise floor so that what is already always the case becomes _____.

4 Spot the slip

Each statement contains a mistake. Correct it in the lines below.

"Someone with a high noise floor simply has a worse mind than someone with a low one."

"The way to quiet mental noise is to fight it: suppress the thoughts and force the mind to be still."

Lesson 3.4: Signal and Noise (continued)

5 Draw it in your own way

Figure 3.4.a. The lesson's caption reads: “Signal-to-noise ratio: the same world reaches a calm mind clearly (high SNR) and a stressed mind buried in noise (low SNR). The difference is noise, not worth.” Draw the same signal in both frames, with the noise each mind adds. At the foot of each frame say what the person mostly perceives.

A calm mind: high SNR

Mostly perceives:

A stressed mind: low SNR

Mostly perceives:

6 Apply

Tomasi sits down in the evening to read. An argument from lunch keeps replaying, and a vague unease has hung about all week. Using the lesson's four kinds of noise, name the two kinds in this scene, and say what the lesson recommends instead of fighting them.

7 Tracker

In 3.3 the equation gained the filter h(t). In the lesson's own notation, write what Lesson 3.4 adds, and how the three malas bias the meeting. End with the equation the lesson's tracker gives.

8 Check my notes

Key points for 3.4 are on the page marked 3.4 at the back of the handbook. There are no marks and no self-rating: compare your notes and decide for yourself.

Lesson 3.5: Attention and the Construction of Experience

Aim: to see attention as active construction: ψ_m, the mind's own wave, with three parameters practice can reach. **Online:** Practice page and understanding check for 3.5 (vedanta.nz, My Journey).

1 In my own words

In three sentences, name the three dials of the attention wave and say what sets or corrects each.

2 Terms to recall

Write each in your own words, then check the key points at the back.

Term	In my own words
The attention wave ψ_m	
Gain G(t)	
Frequency f_m	
Phase φ_m and the standing tilt $\Delta\varphi$	
Bandwidth BW	
Noise floor	
Prediction error ε	
Dhāraṇā	

3 Fill the gaps

What raises your noise floor? _____ attention.

This transition is meditation (dhyāna): the bandwidth is narrow but the engagement is _____.

The first step in yoga is attention-training: not because attention is the highest achievement, but because attention is the _____.

4 Spot the slip

Each statement contains a mistake. Correct it in the lines below.

"When my nervous system is highly activated, my attention must be steady and well focused."

"The narrower my attention, the better I perceive everything around me."

Lesson 3.5: Attention and the Construction of Experience (continued)

5 Draw it in your own way

Figure 3.5.a. The lesson's caption reads: “The attention wave ψ_m has three dials: gain (how strongly it is driven, set by the body and prāṇa), frequency (how well it matches the object, held by manas), and phase (how well aligned the engagement is, corrected by buddhi).” Draw each dial, and under each write one phrase from the lesson for how that dial feels in experience when it is well set.

Gain $G(t) \cdot A_m$	Frequency f_m	Phase φ_m

6 Apply

At a family dinner, Mere's cousin is warm and friendly all evening, but frowns once at something Mere says. On the drive home Mere can think of nothing but the frown. Using the lesson's terms, say what ψ_x was here, where Mere's attention went, and the question the lesson says a trained mind asks. Do not describe yourself.

7 Tracker

Module 3 builds one equation for experience, lesson by lesson. Write what Lesson 3.5 adds, in the lesson's own notation: how it writes ψ_m, and how the attention wave enters the equation. End with the lesson's own tracker line on what the beat and the amplitude depend on.

8 Check my notes

Key points for 3.5 are on the page marked 3.5 at the back of the handbook. There are no marks and no self-rating: compare your notes and decide for yourself.

Lesson 3.6: Resonance, Rhythm and Coherence

Aim: to understand how a system moves from fragmented noise toward clarity through resonance, rhythm and coherence, in physics and in the contemplative sciences. **Online:** Practice page and understanding check for 3.6 (vedanta.nz, My Journey).

1 In my own words

In three sentences, say what each of the three principles does, and why the lesson calls the coherent state a condition of the instrument rather than the recognition of the deepest Self.

2 Terms to recall

Write each in your own words, then check the key points at the back.

Term	In my own words
Resonance	
Rhythm	
Coherence	
Dinacharya	
Trikaraṇa-śuddhi	
Satsaṅga	
Bhāva	
Saṁjñāna Sūkta	

3 Fill the gaps

You cannot avoid resonance. You can only choose which _____ you expose yourself to.

Coherent states are not new properties of the Self being acquired. They are conditions of the _____.

This is coherence: not the absence of thought or feeling or sensation, but their _____ and coordination.

4 Spot the slip

Each statement contains a mistake. Correct it in the lines below.

"Joining any crowd with strong shared feeling can only do me good: a collective field always uplifts."

"A coherent mind is a blank mind, with no thoughts or feelings left in it."

Lesson 3.6: Resonance, Rhythm and Coherence (continued)

5 Draw it in your own way

Figure 3.6.c. The lesson's caption reads: “Coherence: incoherent waves with random phase sum to a modest, scattered result (a bulb), while coherent waves in phase add to a powerful one (a laser).” Draw a few waves and their result in each frame, and under each frame write one example the lesson gives of the same principle in the body or in relationships.

Incoherent: the bulb

Coherent: the laser

6 Apply

Aroha sings in a weekly kīrtan group and notices she is steadier on those evenings. A month later, at a sports final, the crowd around her turns angry and she finds herself shouting with it, then leaves shaken. Using the lesson's section on collective coherence, say what the collective field does in both cases, and what the lesson says collective practice must always be linked with. Do not describe yourself.

7 Tracker

Write what Lesson 3.6 adds to the module's equation, in the lesson's own notation: what happens to n(t), h(t) (and H(f)), G(t), f_m and Δf when coherence is achieved. End with the lesson's own tracker line on what measures the gap.

8 Check my notes

Key points for 3.6 are on the page marked 3.6 at the back of the handbook. There are no marks and no self-rating: compare your notes and decide for yourself.

Lesson 3.7: Body, Prāṇa and the Energetics of Experience

Aim: to see body, breath and prāṇa as the denser end of the mind's own spectrum, taking part in every term of the equation. **Online:** Practice page and understanding check for 3.7 (vedanta.nz, My Journey).

1 In my own words

In three sentences, say why the lesson treats work with the body and breath as a first intervention point, and which term of the equation the body and prāṇa set directly.

2 Terms to recall

Write each in your own words, then check the key points at the back.

Term	In my own words
"All force is prāṇa"	
Ākāśa	
Apāna	
Vyāna	
Udāna	
Samāna	
Low-frequency reference rhythm	
Heart-rate variability (HRV)	

3 Fill the gaps

Before you can steady the mind through attention training, you must stabilize the _____ (the lesson's word for the body and nervous system).

You cannot talk your way out of a conditioning that is stored in the _____.

The variation is not disorder. The variation is _____.

4 Spot the slip

Each statement contains a mistake. Correct it in the lines below.

"Prāṇāyāma is only a warm-up before the real meditation begins."

"Feeling calm and feeling numb come to the same thing: in both, the gain is simply low."

Lesson 3.7: Body, Prāṇa and the Energetics of Experience (continued)

5 Draw it in your own way

Figure 3.7.b. The lesson's caption reads: “The five prāṇas: prāṇa (intake), apāna (release), samāna (assimilation), udāna (expression), and vyāna (circulation), each supporting a function of the inner instrument.” Draw the five as directions of flow, in any shape you like, and beside each write the function of the inner instrument it supports.

Five directions of flow, one energy

6 Apply

Before an important phone call, Tama notices his breath is fast and high in his chest, his shoulders are tight, and every small sound makes him jump. Using the lesson's correspondence of nervous-system states and guṇas, say which state this resembles, what the lesson says happens to G(t), n(t) and f_m in it, and one step from the lesson's practice that he might take. Do not describe yourself.

If you try slow breathing yourself: keep it gentle, with no breath holds, and stop and breathe normally if you feel lightheaded.

7 Tracker

Write what Lesson 3.7 adds to the module's equation, in the lesson's own notation: what the body and prāṇa do to G(t), h(t), n(t) and f_m. End with the lesson's own tracker line on the five prāṇas.

8 Check my notes

Key points for 3.7 are on the page marked 3.7 at the back of the handbook. There are no marks and no self-rating: compare your notes and decide for yourself.

Lesson 3.8: Two Modes of Knowing: Mind as Bridge Between Domains

Aim: to understand two modes of knowing and the mind as the bridge between the temporal and spectral domains. **Online:** Practice page and understanding check for 3.8 (vedanta.nz, My Journey).

1 In my own words

In three sentences, say what the two modes of knowing are and why the Amritabindu Upanishad can call the mind the cause of both bondage and liberation.

2 Terms to recall

Write each in your own words, then check the key points at the back.

Term	In my own words
Difference-based knowing	
Unity-based knowing	
Aparā vidyā	
Parā vidyā	
Temporal domain (kāla-deśa-nimitta)	
Spectral domain	
Saṁyama	
Samāpatti	

3 Fill the gaps

The path is not about destroying the mind or transcending it, but about _____ it.

Of aparā vidyā the lesson says: it cannot reveal what is not an _____.

The wave suffers when it forgets it is _____.

4 Spot the slip

Each statement contains a mistake. Correct it in the lines below.

"To reach unity-based knowing I must give up reasoning: aparā vidyā is the obstacle."

"In samādhi the mind goes blank and disappears."

Lesson 3.8: Two Modes of Knowing (continued)

5 Draw it in your own way

Figure 3.8.a. The lesson's caption reads: “Mind as the bridge between two domains: the temporal domain of separation (aparā vidyā, Δf > 0) and the spectral domain of unity (parā vidyā, Δf → 0).” Draw the two domains and the mind as the bridge between them. On each side write two features the lesson lists for that domain.

Temporal domain	The bridge	Spectral domain

6 Apply

The night before a job interview, Losa lies awake: “What if I freeze? What if they see I am not good enough?” Using the lesson's section on the bridge in mental health, describe in its terms what is happening to the beat, and the small shift the lesson describes for the same situation. What has changed, and what has not? Do not describe yourself.

7 Tracker

Lesson 3.8 has no tracker section. Write what it adds in the module's notation: the value of Δf in each mode of knowing, and what the resultant becomes when Δf = 0. Then write one line on where the lesson places Self-Recognition.

8 Check my notes

Key points for 3.8 are on the page marked 3.8 at the back of the handbook. There are no marks and no self-rating: compare your notes and decide for yourself.

3.1 Vṛtti: The Waves of the Mind

Key points, not model answers. Compare them with your own notes and add what you missed in another colour.

The main ideas

- Yoga Sūtra 1.2, yogaḥ citta-vṛtti-nirodhaḥ: yoga is the stilling of the waves in the mind. The lesson calls this a precise statement, not a metaphor for self-improvement or relaxation.
- A vṛtti (root vṛt, to turn, to whirl) is a wave-formation in the mind-field that arises, has duration and intensity, and subsides. The wave is not separate from the water: a vṛtti is consciousness appearing in a particular form.
- Every perception, thought, emotion, memory and fantasy, even dreamless sleep, is a wave-formation, so all experience can be understood through the physics and mathematics of waves.
- Every wave has two complete descriptions, the temporal domain ψ(t) (sequential) and the spectral domain Ψ(f) (simultaneous). The Fourier transform passes between them with no loss, as the C-major chord shows.
- Ordinary consciousness is temporal-domain consciousness; Consciousness-Energy, described as beyond time, space and causation, the lesson calls a spectral-domain description.
- Swami Vivekananda's lake: the lake is the chitta, the waves the vṛttis, the bottom our own true Self. The Self is not covered as dirt covers a window; it is the substance of which vṛttis are formations. Stilling is the agitation ceasing, as when the wind stops.
- Five types: pramāṇa; viparyaya, which feels exactly like pramāṇa while it lasts; vikalpa, a construct with no external object, not falsehood; nidrā, still a vṛtti; smṛti, revived from saṁskāras and coloured by your present state. Recognising the type is the beginning of freedom.
- ψ(t) = A · sin(2πft + φ): amplitude from the body and prāṇa; the frequency of attention from manas; phase shaped by ahaṁkāra's standing tilt and corrected by buddhi; chitta the field and the record.
- A “fixed” problem is a formation, and formations can change. Observed without identification, vṛttis lose their power, not through suppression but through recognition.

Terms

Citta-vṛtti-nirodha	The stilling of the fluctuations in the mind-stuff; not stopping all awareness.
ψ(t) and Ψ(f)	The same wave described in time and in frequency; both complete.
Viparyaya	A vivid vṛtti that misrepresents what is there (the rope seen as a snake).
Vikalpa	A mental construct with no corresponding external fact: imagination, worry.
Phase φ	In the mind, how aligned and coherent the vṛttis are.

Answers to the exercises

- **Gaps:** pramāṇa; frequency; nothing (the transform is lossless).
- **Slips:** stilling is not destroying the waves or creating a blank mind but the agitation ceasing, so that what was always present reveals itself. Nidrā is one of the five types: minimal content and structure, but still a vṛtti.
- **Figure:** amplitude by the body and prāṇa; frequency by manas; phase by ahaṁkāra's standing tilt and buddhi's correction; duration shaped by the saṁskāras held in chitta.
- **Apply:** vikalpa (worry about what has not happened). She might ask: “What vṛtti is present? What is its amplitude, frequency, and phase?” The worry is a wave-pattern, not who she is.
- **Tracker:** experience is made of vṛttis, each with amplitude, frequency, phase and duration, formed by manas, ahaṁkāra and buddhi in the field of chitta, powered by the body and prāṇa: ψexp(t) = Σ vṛtti (A, f, φ, t).

3.2 Interference: How the Knower Meets the Known

Key points, not model answers. Compare them with your own notes and add what you missed in another colour.

The main ideas

- Superposition: when two waves meet, the medium carries their sum. Nothing new is created: analysed, the sum shows only the two original frequencies, each intact, yet the felt pulse belongs to neither wave alone.
- $\psi exp(t) = \psi_x(t) + \psi_m(t)$. You never experience the bare object: you live the resultant of the object-wave and the attention wave, added in the one field where both arise. That whole resultant R(t) is the vṛtti.
- The carrier, at the mean frequency $(f_s + f_m)/2$, is the content layer, jointly authored by object and mind. The envelope is the clarity layer; its beat is $|f_s - f_m|$: rough and flickering when Δf is large, smooth when small, a constant fullness when $f_m = f_s$. The envelope governs how much of the carrier arrives.
- The clock you did not hear: the sound was fully real, but toward it the attention wave's amplitude was zero, so no vṛtti formed. The object exists without you; the object-as-known co-arises only in the meeting (Pratyabhijñā Hṛdayam, Sūtra 3).
- Ahaṁkāra is not a layer of the resultant but older than it: the standing phase-offset Δφ the attention wave brings to every meeting. With matched frequencies, fullness scales as cos(Δφ/2); at 180 degrees the meeting cancels.
- Superposition in the mathematics, superimposition (adhyāsa) in the philosophy: the felt two-ness is an appearance arising in the meeting. Practice removes the misalignment (Δf → 0, Δφ → 0): $R(t) = 2A \cdot \cos(2\pi f_s t)$, the object alone shining (Yoga Sūtras III.3), the lesson calls this knowing parā vidyā. Self-Recognition lies beyond even this.
- Superposition is a linear law: the medium responds in proportion to what it receives (the “linear zone”). Under stress the mind leaves it and manufactures frequencies of its own. “Gain” is how strongly the attention wave is driven; the remedy the lesson gives is to lower it.
- Manas sets f_m, ahaṁkāra is the standing tilt Δφ, buddhi corrects the phase and reads the carrier, the body modulates amplitude, and chitta is the field in which the superposition happens: all at once, not in a chain.

Terms

Carrier	The content layer: the what of perception, at the mean frequency.
Envelope	The clarity layer: the how of perception, the felt quality of engagement.
Δφ	Ahaṁkāra's standing tilt toward “me and mine”, brought to the meeting.
Adhyāsa	Superimposition: the appearance of separation upon what is not separate.

Answers to the exercises

- **Gaps:** medium; mean; fullness.
- **Slips:** nothing new is created; each wave remains fully itself, and the beat is the pattern of their togetherness. The sound was fully real; no vṛtti formed because the attention wave was elsewhere, so the hearing never came into being.
- **Figure:** a fast oscillation (carrier) whose height swells and fades (envelope); one beat runs from swell to swell.
- **Apply:** ψ_x, the director's note; ψ_m, his expectation and mood. A large Δf (a rough beat) and a Δφ angled toward “What does this say about me?” Before rehearsal: a few slow breaths, a choice to listen rather than defend.
- **Tracker:** the meeting, $\psi exp(t) = \psi_x(t) + \psi_m(t)$; carrier at $(f_s + f_m)/2$, beat at $|f_s - f_m|$; $\Delta f = |f_s - f_m|$; Δφ, fullness as $\cos(\Delta\varphi/2)$; full alignment $R(t) = 2A \cdot \cos(2\pi f_s t)$.

3.3 Saṁskāra as Live Filter

Key points, not model answers. Compare them with your own notes and add what you missed in another colour.

The main ideas

- A tuning fork's impulse response, how it rings and settles, is set by its structure. The mind's impulse response h(t) is set by conditioning: the patterns stored over current and previous lifetimes.
- Three features to notice in lived experience: settling time, overshoot and residual ripple. Learning to notice them is not to blame yourself; it is to wake up to how your mind actually works.
- A well-damped response is proportionate and settles relatively quickly; an under-damped one overshoots, rings and takes longer to settle. Damping comes from ethics and integrity, viveka, breath regulation, a calm environment and unhurried time, and most deeply from understanding your own patterns.
- Convolution, $\psi_x(t) * h(t)$: the incoming signal is filtered by conditioning before attention meets it. The present is smeared by the past, so two people hear the same criticism differently.
- The transfer function H(f), the Fourier transform of h(t), says how much the mind amplifies or suppresses each frequency. It is not unchangeable: practice and viveka reshape it.
- A saṁskāra is stored as a frequency pattern, not at a moment in time; a cue that shares its signature can reconstruct the whole, even when the present situation is quite different.
- Pribram's holonomic model: memory is the interference pattern of the event and your inner state, so event, reaction and meaning return together. To transform it, change the reference beam; there is no location to erase.
- Chitta holds h(t), like a riverbed whose channels can be re-carved. Buddhi can recognise "this is my conditioned version" while the reaction is still felt; the pattern does not disappear in that moment, but something shifts.
- By the convolution theorem the filtering is built into perception, which is why conditioning is invisible. Threat, shame and love responses are different filters; none is right or wrong, and all are teachable.

Terms

Impulse response h(t)	How a system responds to a brief input; for the mind, the filter formed by saṁskāras.
Convolution	The incoming signal combined with the memory of past responses to similar signals.
Transfer function H(f)	The same conditioning seen by frequency: what is amplified, what is suppressed.
Reference beam	In the hologram picture, the inner state an event is recorded against.

Answers to the exercises

- **Gaps:** proportionate; multiplication; reference.
- **Slips:** the reaction depends on the input and on the h(t) it passes through: the criticism plus your history with criticism. A saṁskāra is a distributed frequency pattern, reconstructed by a matching cue, not kept at one time and place.
- **Figure:** well-damped, one smooth rise and fall, short settling time; under-damped, a peak above what is warranted (overshoot), then ringing (ripple) and a longer settling time.
- **Tracker:** h(t), chitta's saṁskāric filter, convolved with the object-wave: $\psi_x(t) * h(t)$, the conditioned signal, which the attention wave then meets. The lesson's equation: $\psi exp(t) = [\psi_x(t) * h(t)] + \psi_m(t)$.

3.4 Signal and Noise: Why the Mind Distorts

Key points, not model answers. Compare them with your own notes and add what you missed in another colour.

The main ideas

- Noise is any unwanted variability that hides a signal of interest. The signal is still there; it is masked, as a distant light is hidden by fog.
- SNR = P_signal / P_noise; in decibels 10, 20 and 30 dB mean ten, a hundred and a thousand times stronger. Different noise floors are a difference in system configuration, not a character flaw: neither person is morally superior or inferior.
- Below the noise floor no signal can reach consciousness. Meditation, breath regulation, ethical living, the repetition of sacred names, silence and simplicity lower it.
- Four kinds of random noise: white (random chatter), pink (persistent moods and background unease), narrowband (the obsessive thought) and impulse (the sudden intrusive spike).
- The three malas are systematic bias, like a camera that tints every photograph: the tint is in the instrument, not the scene. Āṇava mala, the sense of lack, is a distorted standing phase offset Δφ in ahaṁkāra (the nearest analogy is a DC offset). Māyīya mala, the sense of separation, is buddhi's miscalibrated phase correction. Kārma mala, the compulsion to repeat, is a long impulse response h(t) in chitta.
- The ache of each mala is possible only because what it covers is still faintly present. All this noise is in the instrument, not in what you are.
- Patañjali's nine antarāyas (Yoga Sūtra 1.30) and their companions (1.31) describe what noise looks like in daily life; the remedy is eka-tattva-abhyāsa (1.32), holding the attention wave on one object. Mantra, breath-awareness or one chosen ideal give a stable reference rhythm.
- Lowering the noise floor works better than struggling against the noise, which is like turning up the radio to reduce static.
- Practice does not manufacture peace or the witness. As the floor drops, peace and then the space between thoughts become perceptible; then you recognise yourself as the witness (sākṣī), not as a new object; at the deepest stillness the Ātman recognises the Ātman. Nothing is added, only unveiled.

Terms

Noise floor	The baseline level of noise below which signals cannot be detected.
Mala	A systemic distortion that colours everything processed; not a moral judgment.
Eka-tattva-abhyāsa	Sustained practice of concentration on one principle (Yoga Sūtra 1.32).

Answers to the exercises

- **Gaps:** hundred; impulse response (h(t)); manifest.
- **Slips:** neither has a better or worse mind; they have different noise floors, a difference in configuration, not worth. Fighting noise is like reducing static by turning up the volume; lower the baseline through systemic practices instead.
- **Figure:** the same signal in both; in the calm mind it comes through clearly, in the stressed mind it is buried and the person experiences mostly the noise.
- **Apply:** narrowband (the replayed argument) and pink (the background unease). Lower the floor: simplify input, regulate the nervous system, live ethically, repeat what is wholesome.
- **Tracker:** n(t), the unwanted noise. The three malas bias the meeting: āṇava a distorted Δφ, māyīya buddhi's miscalibrated correction, kārma a long h(t). The lesson's tracker (§13): $\psi exp(t) = [\psi_x(t) * h(t)] + \psi_m(t) + n(t)$.

3.5 Attention and the Construction of Experience

Key points, not model answers. Compare them with your own notes and add what you missed in another colour.

The main ideas

- Attention IS ψ_m: $\psi_m(t) = A_m \cdot \cos(2\pi f_m \cdot t + \varphi_m)$. It enters the equation scaled by a slowly varying gain G(t), so that what meets the object is $G(t) \cdot \psi_m(t)$.
- Gain is set by the body and prāṇa (arousal, sleep, nourishment, breath, health) and trained indirectly through them. High arousal can come with scattered attention, low arousal with steady attention: the product $G(t) \cdot A_m$ matters.
- Frequency f_m is held by manas. When f_m matches f_s, the beat $\Delta f = |f_s - f_m|$ shrinks. Phase is not manas's work: the attention wave arrives with a standing tilt $\Delta\varphi$, which the ahaṁkāra carries, and buddhi corrects it. Phase-locking is felt as ease.
- Bandwidth: BW · Δt ≥ constant. Narrow attention (dhāraṇā) knows one thing deeply but misses change; wide attention stays responsive but shallow. In dhyāna the bandwidth is narrow but the engagement effortless.
- Scattered attention raises the noise floor; focused, sustained attention lowers it, and signals that were always there can be detected.
- Predictive processing: $\varepsilon(t) = \psi_x(t) - \psi_predicted(t)$. The anxious mind lives mostly its prediction (high w_1); the trained mind notes the prediction, holds it lightly and asks, "what is actually present right now?" (high w_2).
- Manas deploys attention; buddhi guides it. A mind focused on its own conditioned beliefs "is not more free; it is more imprisoned". Viewed through this lens, anxiety and depression are attention-allocation problems; throughout, the deepest Self is not damaged.

Terms

ψ_m	The attention wave: the mind's own wave of engagement, with amplitude, frequency and phase.
G(t)	The gain: the strength with which the body and prāṇa drive the attention wave.
f_m; φ_m and $\Delta\varphi$	How well the mind's rhythm matches the object (manas); how well aligned the engagement is (buddhi corrects ahaṁkāra's standing tilt $\Delta\varphi$).
ε; w_1, w_2	The prediction error, which carries genuinely new information; the weights given to prediction and to prediction error.

Answers to the exercises

- **Gaps:** scattered; effortless; gate.
- **Slips:** an anxious person can be scattered despite being activated, and a tired person can stay focused; steadiness is f_m, held by manas. Narrow bandwidth gives depth but you become insensitive to change (BW · Δt ≥ constant).
- **Figure task:** gain ("intense and clear"), frequency ("the flicker disappears"), phase ("a felt rightness, an ease").
- **Apply:** ψ_x is the whole, mostly friendly evening; attention locked onto one moment, so that became the experience. The question: "what is actually present right now?"
- **Tracker:** $\psi_m(t) = A_m \cdot \cos(2\pi f_m \cdot t + \varphi_m)$, entering as $G(t) \cdot \psi_m(t)$. "The beat frequency depends on f_m, which manas sets and attention-training steadies. The amplitude of the vṛtti depends on G(t), which the body and prāṇa set and which is trained indirectly through them."

3.6 Resonance, Rhythm and Coherence

Key points, not model answers. Compare them with your own notes and add what you missed in another colour.

The main ideas

- Resonance: when a driving frequency matches a system's natural frequency, energy transfers with maximum efficiency (tuning forks, soldiers breaking step on bridges, radio, atoms). In the mind it works in both directions: steady and peaceful, or agitated and fearful.
- Q measures how selective and sharp a resonance is; dhāraṇā, dhyāna and viveka work to increase it. Q is neutral in itself: a trained mind's high Q is selectivity of attention, not lingering reactivity.
- Rhythm is the stabiliser: heartbeat, breath, circadian and seasonal cycles. Dinacharya, a regular daily routine, lets the nervous system settle; japa, posture and breath regulation create rhythm.
- Read as rhythm, sattva is rhythmic coherence, rajas irregular over-activity, and tamas rhythmic collapse: “the stillness of inertia, not rest”.
- Coherence is the degree to which the parts of a system are in phase. Bulb and laser: same energy, utterly different effect. Trikaraṇa-śuddhi is the unity of thought, word (or feeling) and action. Dhāraṇā is the beginning of coherence; dhyāna is coherence stabilised.
- In the coherent state n(t) drops, h(t) becomes well-damped (the past is not erased; H(f) approaches 1), G(t) steadies, and f_m approaches f_s, so Δf shrinks toward zero: “what sattva looks like in equation form”. It is not the recognition of the deepest Self but the condition of the instrument that allows it.
- Satsaṅga: the lesson describes two coupled nervous systems tending to synchronise; choosing companions and environment is a fundamental practice. Words of a teacher, texts and chants can also help. Bhāva is a sustained, cultivated feeling-tone, not positive thinking.
- The Saṁjñāna Sūkta (Ṛgveda X.191.2 to 4) names moving, speaking, knowing, feeling and intending together. The collective field amplifies whatever is present, so collective practice is linked with viveka.

Terms

Resonance	Frequency match producing maximum energy transfer.
Coherence	Parts in phase, their oscillations aligned and coordinated.
Trikaraṇa-śuddhi	“Purity and unity of the three instruments”: thought, word (or feeling) and action.
Bhāva	Feeling-tone: a sustained, cultivated feeling-state, rehearsed until it becomes natural.

Answers to the exercises

- **Gaps:** frequencies; instrument; alignment.
- **Slips:** the collective field is neutral and amplifies whatever frequency is present, fear or rage included. Coherence is not the absence of thought, feeling or sensation but their alignment; sattva is not being completely still.
- **Figure task:** random phases sum to a modest average; waves in phase add. Examples: breath and heartbeat coordinated; words matching feelings.
- **Apply:** in both cases the field amplifies the frequency present: steadiness in the kīrtan, anger in the crowd. Collective practice is linked with discernment (viveka).
- **Tracker:** n(t) drops; h(t) well-damped, H(f) → 1; G(t) steady; f_m approaches f_s; Δf shrinks. Of the ordinary and coherent states: “The gap between these two equations is the gap between bondage and clarity. It is measured by Δf.”

3.7 Body, Prāṇa and the Energetics of Experience

Key points, not model answers. Compare them with your own notes and add what you missed in another colour.

The main ideas

- Swami Vivekananda, in Raja Yoga: all force is prāṇa. Ākāśa is the passive principle; prāṇa the dynamic, manifesting power. Prāṇāyāma is not a preliminary exercise but direct engagement with that principle.
- The five prāṇas are five directions of flow of one energy: prāṇa powers manas's reception of ψ_x; apāna supports chitta in releasing, shortening the ringing of h(t); vyāna carries the signal across the inner instrument; udāna supports the move from interference to resonance, f_m drawn toward f_s; samāna supports buddhi's assimilation of the carrier into meaning.
- Breath is a low-frequency reference rhythm. Slowed to about six breaths a minute (0.1 Hz), the lesson says, it gives the system a stable reference, reduces n(t), stabilises G(t) and steadies f_m, so Δf is no longer erratic. Hence prāṇāyāma precedes dhāraṇā: first stabilise the hardware.
- The lesson's correspondence, drawing on Stephen Porges's polyvagal theory (“vagal” relates to the vagus nerve, which helps regulate the body's calm and alert states): ventral vagal with sattva (low noise floor, stable G, coherent frequencies); sympathetic with rajas (high gain, raised noise floor, erratic f_m); dorsal vagal with tamas (G very low, sluggish f_m, heavy and weak processing).
- The body takes part in every term. Muscular armour, posture and breath patterns are recorded as saṁskāras in chitta and so become part of h(t). Pain, fatigue, hunger and background arousal are noise. Sympathetic activation raises gain (A_output = G · A_input). The attention wave tends to fall into step with the body's dominant rhythm.
- Body, breath and prāṇa are Consciousness-Energy in Its densest mode. The body is not a distraction from inner work; it is the entry point.

Terms

Prāṇa (Swami Vivekananda)	The active principle of the universe: the infinite, omnipresent manifesting power.
Apāna; vyāna	Release and letting go; circulation and distribution across the whole system.
Udāna; samāna	Expression and growth; digestion and assimilation of experience into understanding.
Heart-rate variability (HRV)	The natural beat-to-beat variation in the time between heartbeats; the lesson says the variation is health.
Reference rhythm	Slow breath as a stable low-frequency reference: like a tuning fork held steady, it gives the system something to synchronize with.

Answers to the exercises

- **Gaps:** hardware; body; health.
- **Slips:** Swami Vivekananda insisted prāṇāyāma is not a warm-up but direct engagement with the cosmic energy principle. In the lesson's correspondence, calm (ventral vagal) goes with stable gain and shutdown (dorsal vagal) with very low G, so experience becomes muffled and distant.
- **Figure task:** intake: manas's reception; release: chitta; circulation: the whole instrument; expression: f_m toward f_s; assimilation: buddhi.
- **Apply:** sympathetic activation, which the lesson maps to rajas: G(t) shoots up, n(t) rises, f_m becomes erratic. A step: slow the breath gently (four in, six out), or posture, shoulders dropped and chest open.
- **Tracker:** G(t) set by the body and prāṇa through the autonomic state; h(t) includes somatic conditioning; n(t) includes physiological noise; f_m influenced by breath rhythm. “The five prāṇas flow through all of this. Regulate them through conscious breathing, posture, and body awareness, and you regulate the entire system from the ground up.”

3.8 Two Modes of Knowing: Mind as Bridge Between Domains

Key points, not model answers. Compare them with your own notes and add what you missed in another colour.

The main ideas

- “Mind alone is the cause of both bondage and liberation for human beings” (Amritabindu Upanishad). The mind is a neutral instrument with two modes; the path is retuning it.
- The temporal domain (kāla-deśa-nimitta: time, space, cause) is sequential, with subject and object separate: difference-based knowing, aparā vidyā. The spectral domain, Consciousness in its native mode (cit-prakāśa), is whole and simultaneous: unity-based knowing, parā vidyā, the knowing by which the Imperishable is attained (Muṇḍaka 1.1.5).
- The mind is the bridge between the two domains. Difference-based mode: $\Delta f > 0$ and a beat, the felt gap between knower and known. Unity-based mode: $\Delta f \to 0$, the beat slows and stops; at $\Delta f = 0$ the resultant is a single wave at the object's own frequency, doubled in fullness, the standing tilt released. Whether this is a literal mechanism in the brain is, the lesson says, an open scientific question.
- Dhāraṇā, dhyāna and samādhi progressively reduce the beat until unity-based knowing emerges naturally: this is saṁyama. In samādhi, $\Delta f = 0$.
- Aparā vidyā (measurement, reasoning, perception, memory) is essential and not to be rejected, but cannot reveal what is not an object. Parā vidyā is not opposed to logical thinking but prior to it.
- The lesson reads anxiety as the beat locked open, rumination and depression as the beat heavy and stuck, trauma as the bridge strained (with skilled support). The mind stops being the enemy.
- AI is aparā vidyā at scale: it can simulate unity-based knowing, not be it. YS 1.41: the transparent crystal-mind takes on the form of knower, knowing and known (samāpatti).
- “When $\Delta f = 0$, the beat stops completely: this is parā vidyā, the perfected resultant, samādhi. Self-Recognition goes one step further, and belongs not to the equation but to the one who sees it.”

Terms

Aparā vidyā	Lower or worldly knowledge: difference-based, all forms of “knowledge about”.
Parā vidyā	Higher knowledge: unity-based, the knowing by which the Imperishable is attained.
Saṁyama	Dhāraṇā, dhyāna and samādhi as one practice that reduces the beat.
Samāpatti	Absorption: knower, knowing and known one in the crystal-mind (YS 1.41).

Answers to the exercises

- **Gaps:** retuning; object; ocean.
- **Slips:** aparā vidyā is essential; parā vidyā is prior to logic, not opposed to it. In samādhi the mind does not disappear: like a transparent crystal it becomes the perfect instrument.
- **Figure task:** temporal: sequential, subject and object separate, many parts. Spectral: whole, simultaneous, timeless, the one appearing as many.
- **Apply:** the beat locked open, Δf wide. The shift: three slow breaths and noticing, “These are thoughts arising in awareness.” The thoughts remain; the relationship has shifted and the beat loosens.
- **Tracker:** difference-based, $\Delta f > 0$; unity-based, $\Delta f \to 0$; at $\Delta f = 0$ a single wave at the object's own frequency. Self-Recognition goes beyond parā vidyā, and belongs not to the equation but to the one who sees it.